

Jiacheng Lai

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EDUCATION

School of Mathematics | Changsha University
B.S. in Mathematics and Applied Mathematics

Sept. 2023 - Present

EMPLOYMENT

Shenzhen University of Advanced Technology
Research Intern | Advisor: Hui Peng

Jul. 2026 - Present

Bioinformatics Complex Network Big Data Research Institute | Changsha University
Undergraduate Research Assistant | Advisor: Lei Wang

Oct. 2023 - Jul. 2026

RESEARCH INTERESTS

My research focuses on reliable and calibrated machine learning for computational biology and drug discovery. I am interested in heterogeneous biomedical data integration, uncertainty quantification, low-data prediction, and reliable inference from noisy or incomplete data. My work spans graph-based microbe-disease and microbe-drug association prediction, cold-start drug-microbe learning, and multimodal drug-Mtb affinity prediction for anti-tuberculosis drug repurposing.

PUBLICATIONS AND MANUSCRIPTS

(* denotes equal contribution.)

- [1] (*Journal of Molecular Biology* 2025) **Jiacheng Lai**, Zhen Zhang, Bin Zeng, and Lei Wang. "GTFKAN: A Novel Microbe-Drug Association Prediction Model Based on Graph Transformer and Fourier Kolmogorov-Arnold Networks." [Paper](#) | [Code](#)

TL;DR | Developed GTFKAN, combining Graph Transformer Networks and Fourier Kolmogorov-Arnold Networks to integrate heterogeneous network structure and high-order latent features.

- [2] (*Under Review at Journal of Biomedical Informatics*) **Jiacheng Lai**, Zhen Zhang, Hao Yuan, and Lei Wang. "BMFN: Multimodal Feature Fusion Network for Drug-Mtb Affinity Prediction and Anti-Tuberculosis Drug Repurposing." [Code](#)

TL;DR | Developed a multimodal network integrating sequence, heterogeneous graph, and molecular fingerprint features for anti-tuberculosis drug repurposing.

- [3] (*Submitted to IEEE BIBM 2026*) **Jiacheng Lai***, Xingye Xu*, Zhen Zhang, Zhenyu Yang, Hao Yuan, and Lei Wang. "MiDSFN: A Novel Signature Fusion Learning Network for Potential Drug-Microbe Associations Prediction." [Code](#)

TL;DR | Developed a feature-independent multimodal framework for cold-start drug-microbe association prediction, cross-dataset transfer, and robust evaluation with unverified negatives.

- [4] (*Algorithms* 2026) Zhen Zhang, Xianjun Hu, **Jiacheng Lai**, and Lei Wang. "A Graph Convolutional Network Framework Integrating Nuclear Norm Minimization and Conditional Random Fields for Microbe-Disease Association Prediction." [Paper](#)

TL;DR | Developed NNGCFCAE, integrating nuclear norm minimization, a CRF-enhanced graph convolutional network, and a residual multi-channel convolutional autoencoder for microbe-disease association prediction.

- [5] (*Submitted to BMC Bioinformatics*) **Jiacheng Lai***, Xiangyi Wang*, and Hui Peng. "SLF-MTF: Disease-Derived Pseudo-Associations and Multi-Scale Topological Fusion for Microbe-Drug Association Prediction."

TL;DR | Developed disease-derived pseudo-association inputs and multi-scale graph Transformer fusion to prioritize candidate microbe-drug associations without using known associations as input edges.

PROJECTS

**Project Lead, A Novel Combined Neural Network Model for Predicting
Microbe-Drug Associations Based on Bilinear Attention Networks**

Nov. 2024 - Jun. 2025

*National Undergraduate Innovation and Entrepreneurship Training Program | Project No.
S202411077005*

AWARDS

Second Prize, National College Student Mathematics Competition (Mathematics Group B)	Dec. 2024
Second Prize, Hunan Provincial College Student Mathematics Competition (Mathematics Group B)	Dec. 2024
Honorable Mention, COMAP MCM/ICM	Feb. 2025
Second Prize, Chia Tai Cup National College Student Market Survey and Analysis Competition	Apr. 2025
Third Prize, Challenge Cup Hunan Provincial Competition (Extracurricular Academic Works)	Jun. 2025
Third Prize, Hunan Provincial College Student Market Survey and Analysis Competition	Aug. 2025
Third Prize, Changsha College Student Data Elements-Empowered AI Innovation and Entrepreneurship Competition	Dec. 2025